

## LAB EXERCISE 20

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**B2 DEVOPS**

**APP DEPLOYMENT USING ECS & ECR**

### STEP 1: CREATE ECR REPOSITORY

- GO TO AWS CONSOLE
- SEARCH **ELASTIC CONTAINER REGISTRY (ECR)**
- CLICK **CREATE REPOSITORY**
- GIVE NAME (EXAMPLE: MY-WEB-APP)
- CLICK **CREATE**

The screenshot shows the AWS Management Console interface for creating a new private repository in Amazon ECR. The breadcrumb navigation at the top indicates the path: Amazon ECR > Private registry > Repositories > Create private repository. The page title is 'Create private repository'.

**General settings**

**Repository name**  
Enter a concise name. Repositories support namespaces, which you can use to group similar repositories.  
099695493670.dkr.ecr.ap-south-1.amazonaws.com/ devops-repo  
11 out of 256 characters maximum (2 minimum). The name must start with a letter and can only contain lowercase letters, numbers, and special characters -./.

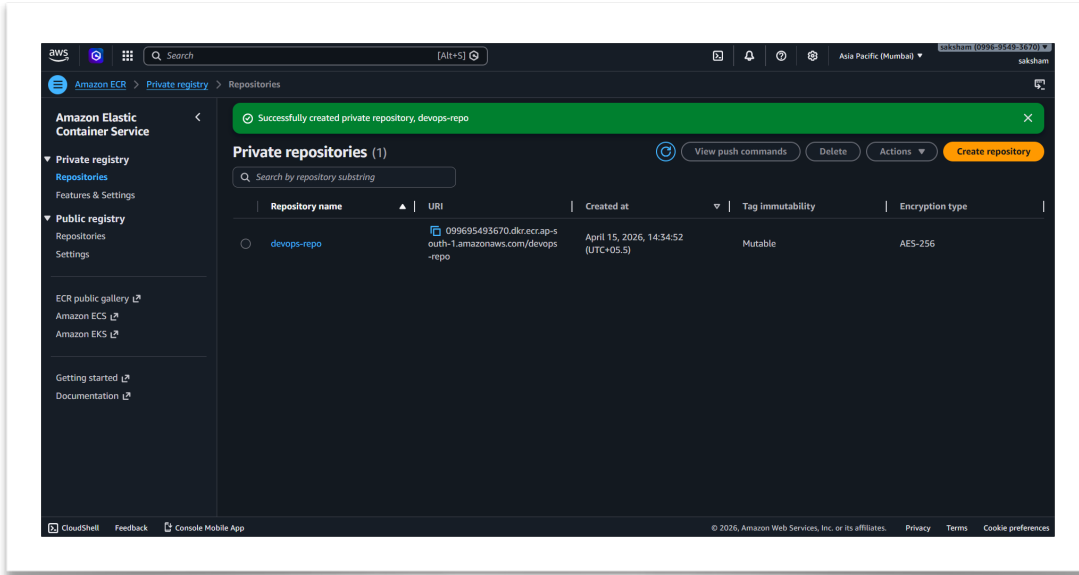
**Image tag settings** [Info](#)

**Image tag mutability**  
Choose the tag mutability setting.  
☒ **Mutable**  
Image tags can be overwritten.  
☐ **Immutable**  
Image tags can't be overwritten.

**Mutable tag exclusions**  
Tags that match these filters will be immutable (can't be overwritten). Using wildcards (\*) will match zero or more image tag characters.  
Add filter  
Filters must only contain letters, numbers, and special characters (-, \*, \_). Each filter is limited to 128 characters, 2 wildcards (\*), and you can add up to 5 filters in the exclusion list.

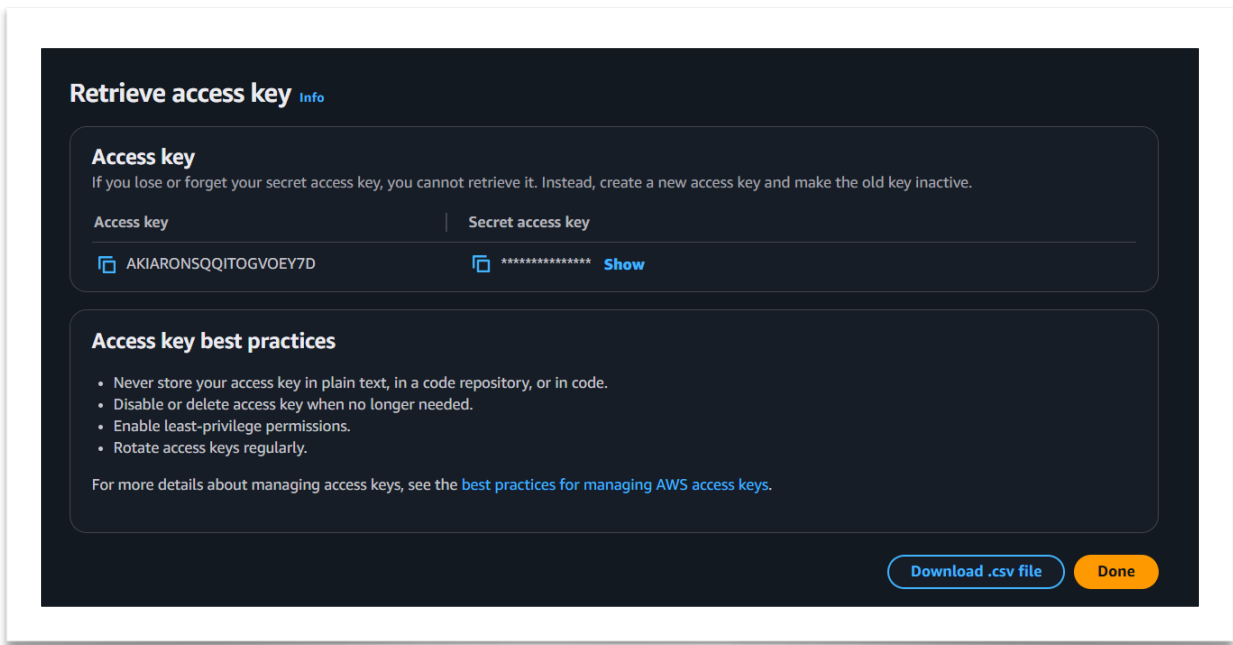
**Encryption settings** [Info](#)

The encryption settings for a repository can't be changed once the repository is created.



## STEP 2: CONFIGURE AWS

### CREATE ACCESS KEY



```
C:\Users\hp>aws configure
AWS Access Key ID [*****SR2E]: AKIARONSQITOGVOEY7D
AWS Secret Access Key [*****I49u]: Sgq3XFduh1KzOpMj2rdilG7fS0bn5dIet9eyRJzT
Default region name [us-east-1]:
Default output format [json]:

C:\Users\hp>
```

### STEP 3: AUTHENTICATE DOCKER TO ECR

RUN THIS :

```
AWS ECR GET-LOGIN-PASSWORD --REGION AP-SOUTH-1 | DOCKER LOGIN --USERNAME AWS --PASSWORD-
STDIN 099695493670.DKR.ECR.AP-SOUTH-1.AMAZONAWS.COM
```

```
C:\Users\hp>aws ecr get-login-password --region ap-south-1 | docker login --username AWS --password-stdin 099695493670.dkr.ecr.ap-south-1.amazonaws.com
Login Succeeded

C:\Users\hp>
```

### STEP 4: BUILD DOCKER IMAGE

RUN THIS :

```
DOCKER BUILD -T DEVOPS-ECR .
```

```
[+] Building 38.2s (9/9) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 93B                                0.2s / 0.0s
=> [internal] load metadata for docker.io/library/httpd:latest    10.0s
=> [auth] library/httpd:pull token for registry-1.docker.io       0.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [1/3] FROM docker.io/library/httpd:latest@sha256:bdba5c8602f2d6ad0783168b07fe98541fe45e97b434b13ba8c959a5850fb6b 26.1s
=> => resolve docker.io/library/httpd:latest@sha256:bdba5c8602f2d6ad0783168b07fe98541fe45e97b434b13ba8c959a5850fb6b 0.4s
=> => sha256:bdba5c8602f2d6ad0783168b07fe98541fe45e97b434b13ba8c959a5850fb6b 18.1MB / 10.1MB 0.0s
=> => sha256:6d1468ced9d5fca3209174032266770bc0d0fca3dcfdce81ee70a337f835d80b 2.89MB / 2.89MB 0.0s
=> => sha256:e48c16e072a6f6e551a262fecb99d54ba13974acc2112d3c683665be2deafc 7.99kB / 7.99kB 0.0s
=> => sha256:5435b2dcd75cb7faa0d5b1d4d54be2c72a776fab9a605336f5067d6e9ecb5976 29.78MB / 29.78MB 22.7s
=> => sha256:8261ed062f8c1c85c3e4b2ae3a9ae5a586b94fcc61b68c110ee65c9c523b49c7 146B / 146B 8.0s
=> => sha256:4f4fb70be540461cfa02571ae0db9a0dc1e0cd5577484a6d75e68dc38e8acc1 52B / 52B 7.6s
=> => sha256:a838a4be55ced5f2d7288c56a3a13ecbd15118574ee338337c50737ae163f4f 2.88MB / 2.88MB 11.6s
=> => sha256:61e1798e23ba12f14505e225391338db290ee6cecfb6321c6c6b92ac8b546dcb 13.45MB / 13.45MB 24.9s
=> => sha256:4e51448b39490b8604d7662b95848e89a3563fa3b995159f7ec54d4d48a4493f 289B / 289B 12.7s
=> => extracting sha256:6d1468ced9d5fca3209174032266770bc0d0fca3dcfdce81ee70a337f835d80b 1.6s
=> => extracting sha256:8261ed062f8c1c85c3e4b2ae3a9ae5a586b94fcc61b68c110ee65c9c523b49c7 0.0s
=> => extracting sha256:4f4fb70be540461cfa02571ae0db9a0dc1e0cd5577484a6d75e68dc38e8acc1 0.0s
=> => extracting sha256:a838a4be55ced5f2d7288c56a3a13ecbd15118574ee338337c50737ae163f4f 0.2s
=> => extracting sha256:61e1798e23ba12f14505e225391338db290ee6cecfb6321c6c6b92ac8b546dcb 0.0s
=> => extracting sha256:4e51448b39490b8604d7662b95848e89a3563fa3b995159f7ec54d4d48a4493f 0.0s
=> [internal] load build context
=> => transferring context: 554B                                0.0s
=> [2/3] WORKDIR /usr/local/apache2/htdocs/                    0.2s
=> [3/3] COPY . /usr/local/apache2/htdocs/                     0.1s
=> => exporting to image                                         0.1s
=> => exporting layers                                           0.1s
=> => writing image sha256:af22f887c7af5788a8ae5e6448980692dbd6e836ebafbea20d4ccc28b819687 0.0s
=> => naming to docker.io/library/devops-ecr                    0.0s
```

### STEP 5: TAG DOCKER IMAGE

RUN THIS :

```
DOCKER TAG DEVOPS-ECR:LATEST 099695493670.DKR.ECR.AP-SOUTH-1.AMAZONAWS.COM/DEVOPS-
```

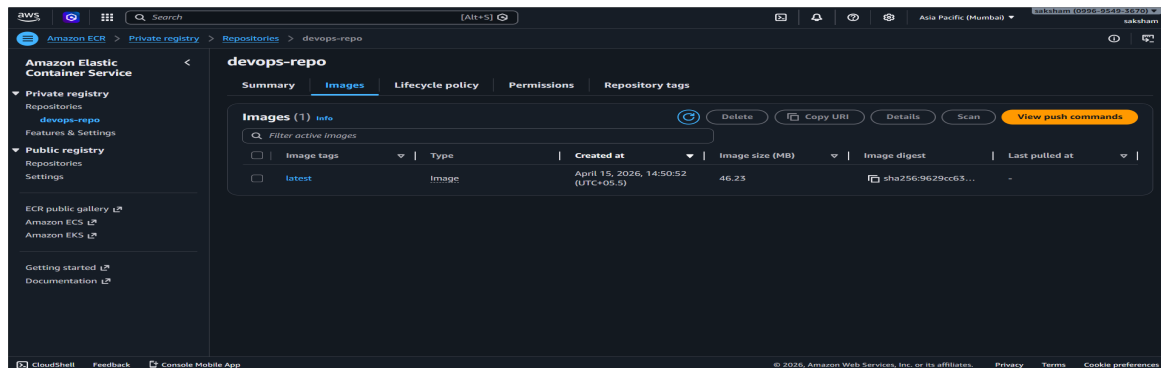
ECR:DEMO

## STEP 6 : PUSH DOCKER IMAGE

RUN THIS :

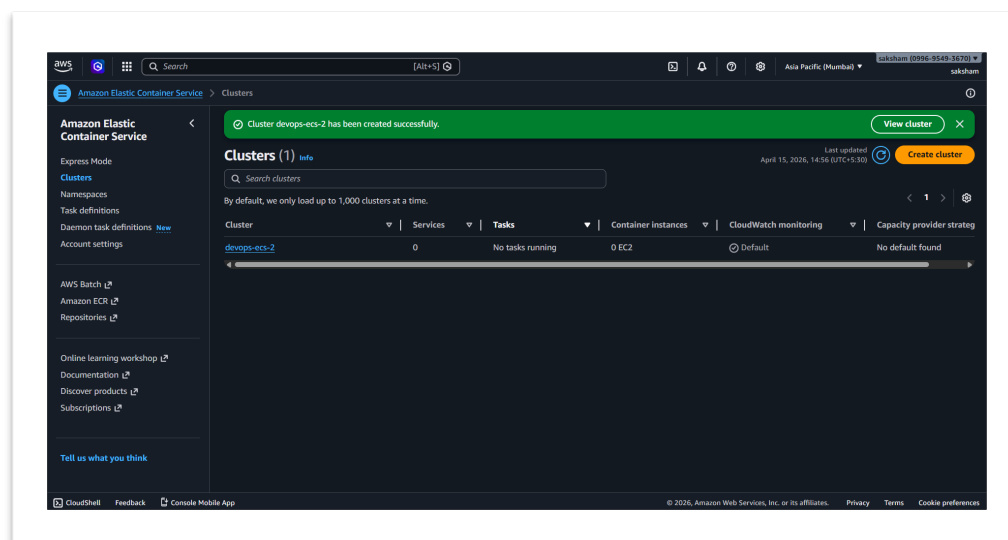
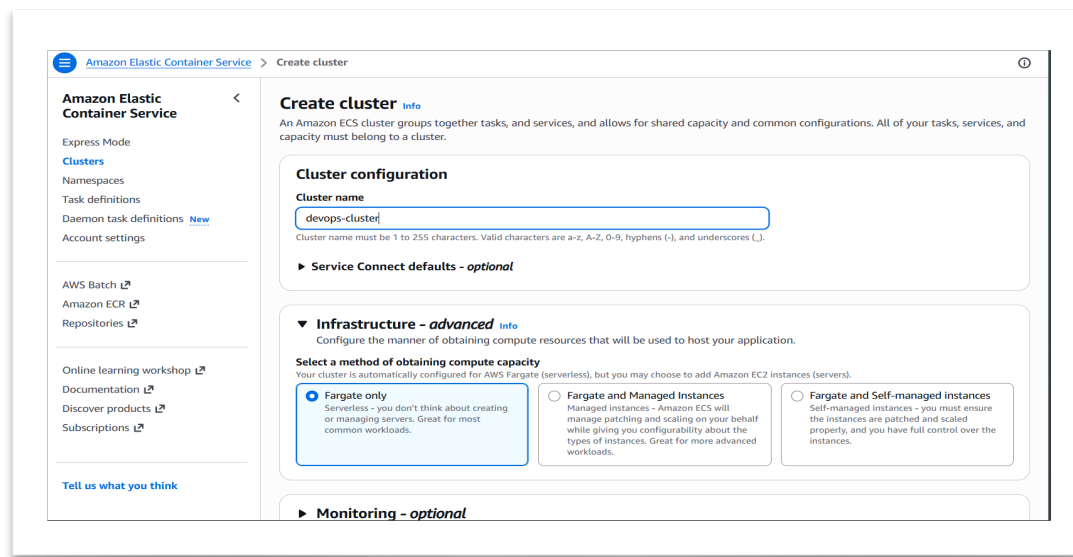
```
DOCKER PUSH 099695493670.DKR.ECR.AP-SOUTH-1.AMAZONAWS.COM/DEVOPS-REPO:LATEST
```

```
The push refers to repository [099695493670.dkr.ecr.ap-south-1.amazonaws.com/devops-repo]
a077f74b3497: Pushed
5f70bf18a886: Pushed
f70b8cc6a4e9f: Pushed
088368c64fc8: Pushed
01708522c61e: Pushed
b5d4ea14e746: Pushed
60e70dddd9ea: Pushed
latest: digest: sha256:9629cc6379fc2d427b0b0aed6ddb4847e7f30d52fca719a403258c48aa8a93f3 size: 1985
```



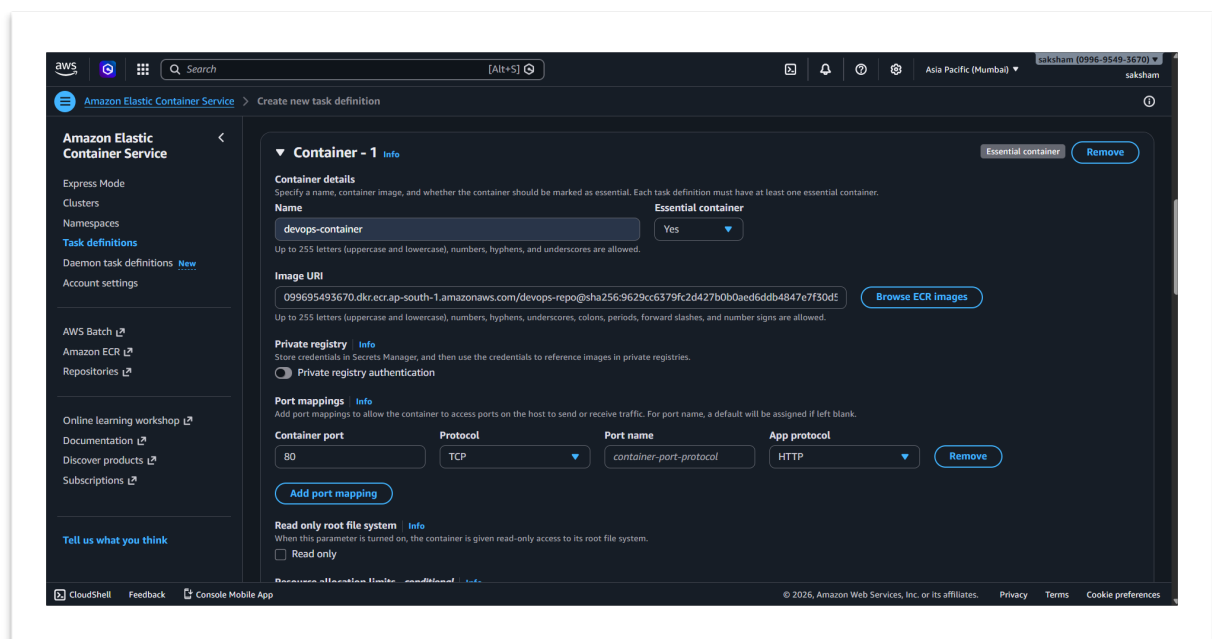
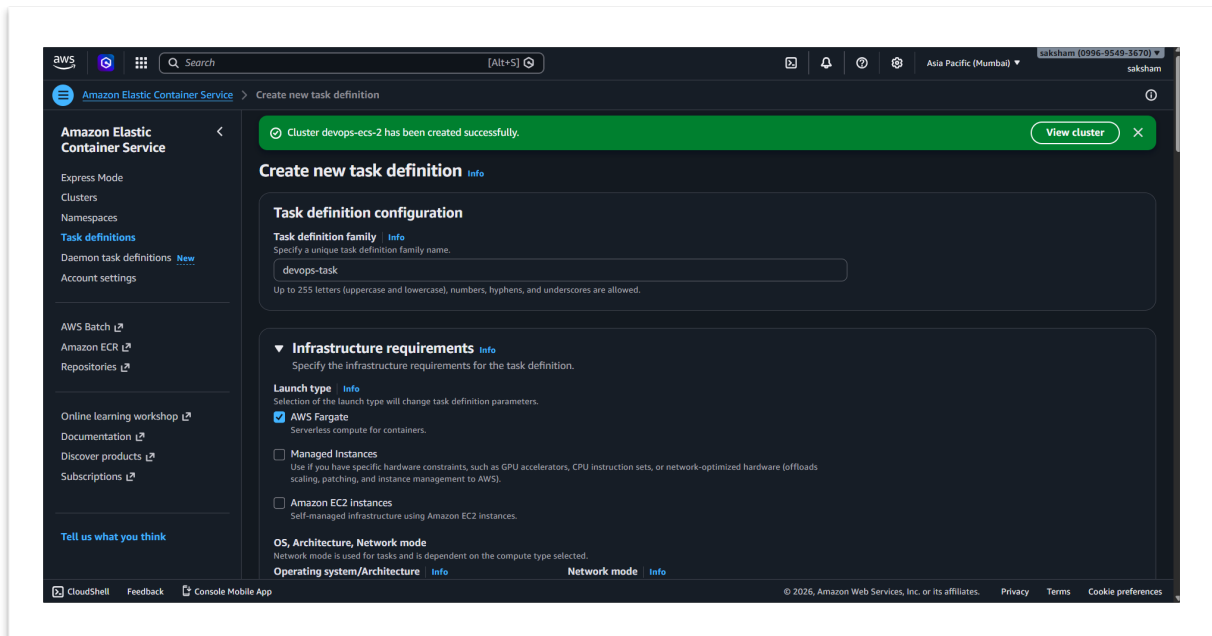
## STEP 7: CREATE CLUSTER

1. GO TO ECS
2. CLICK **CREATE CLUSTER**
3. SELECT **FARGATE (SERVERLESS)**
4. CLUSTER NAME: DEVOPS-CLUSTER-NEW
5. CLICK CREATE



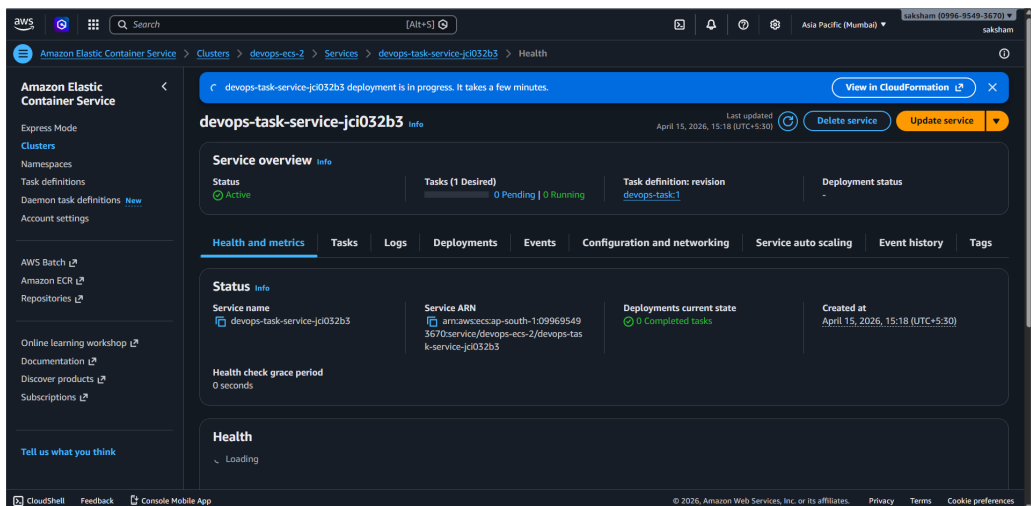
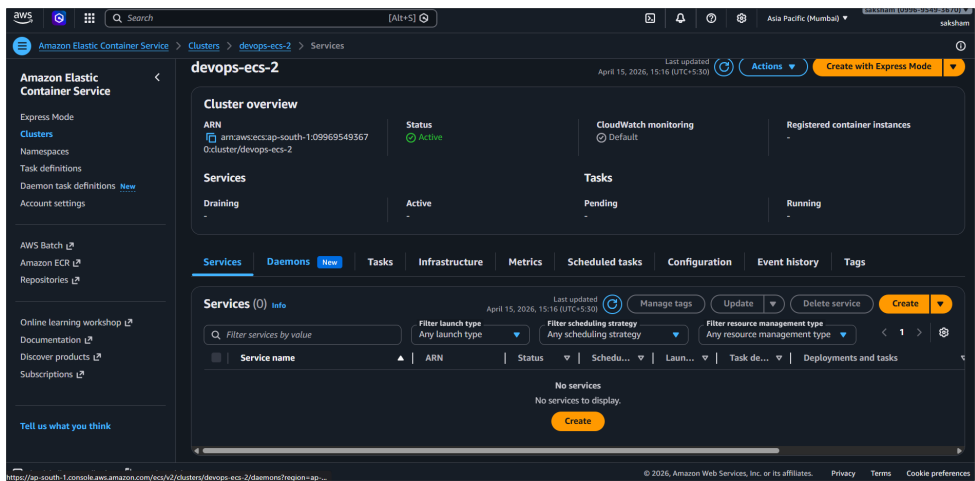
## STEP 8: CREATE TASK DEFINITION

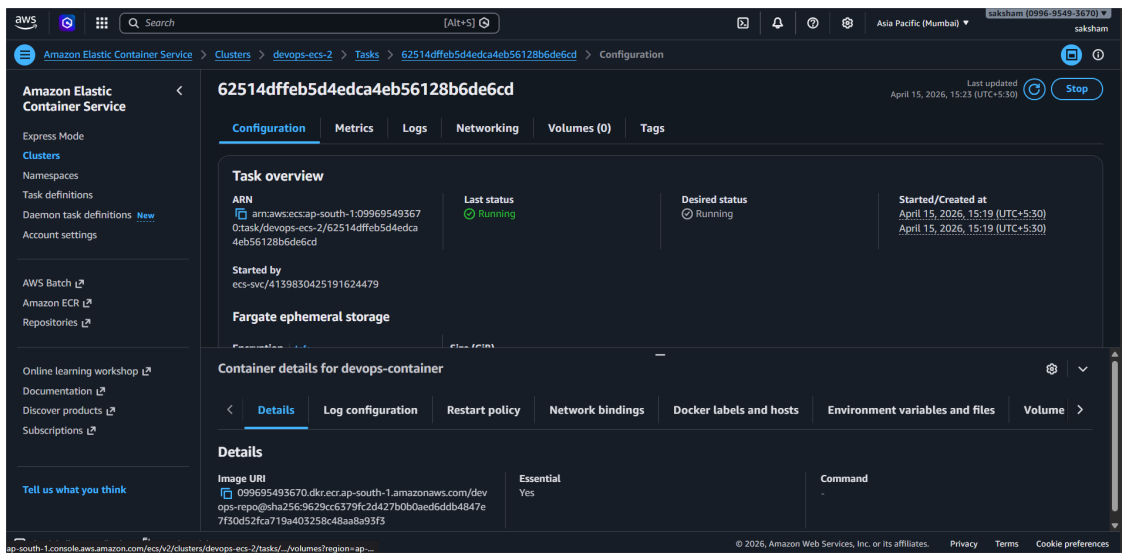
1. LAUNCH TYPE → FARGATE
2. CONTAINER IMAGE → PASTE ECR URI



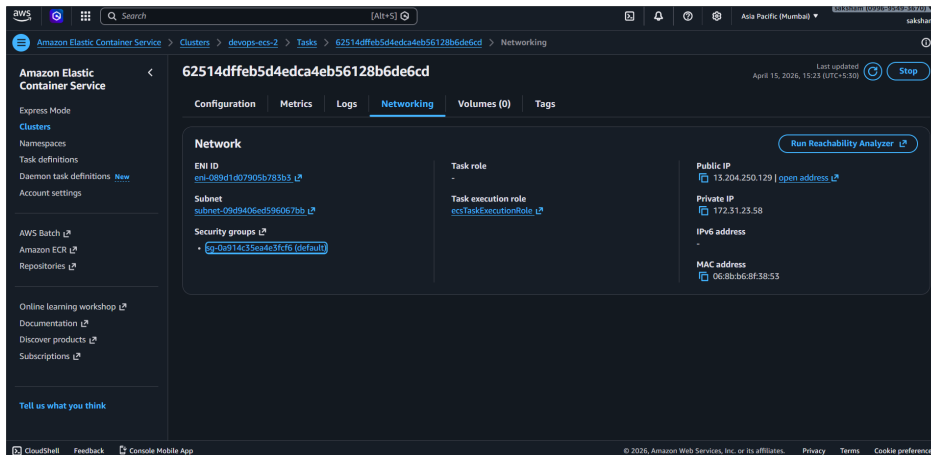
## STEP 9: CREATE SERVICE

1. GO TO CLUSTER
2. CLICK **CREATE SERVICE**

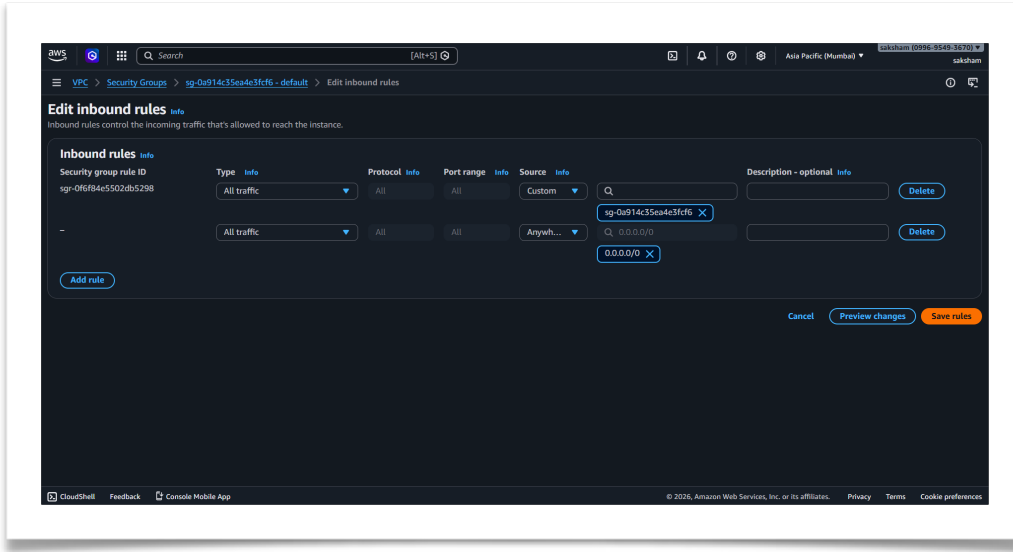




## STEP 10: TURN ON ALL TRAFFIC IN SECURITY BROKE



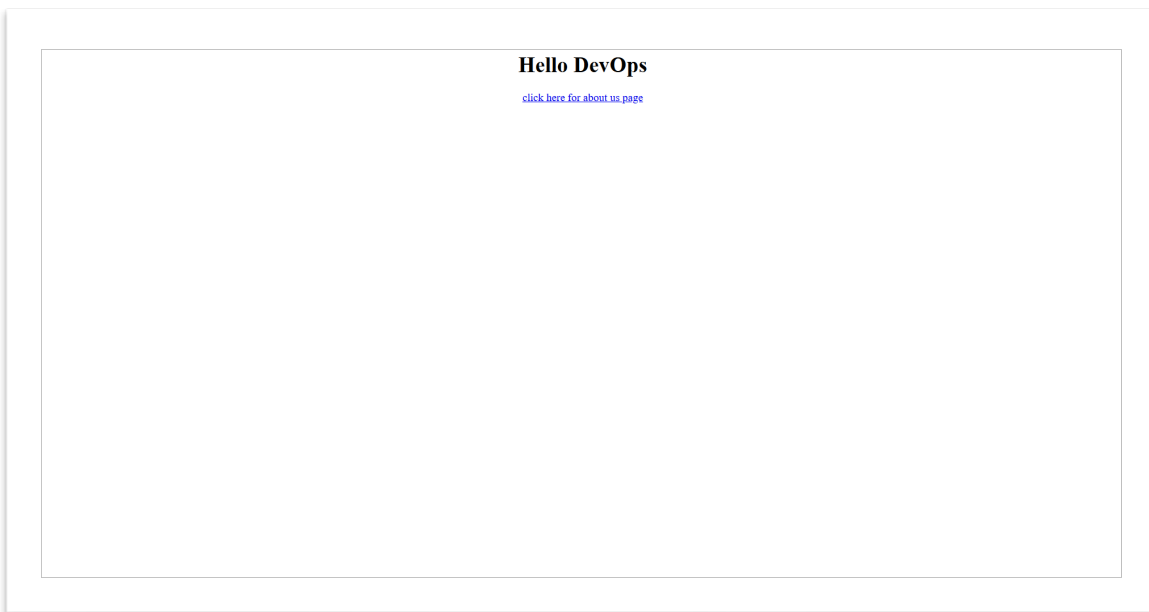




## STEP 11 : VERIFY DEPLOYMENT

1. GO TO LOAD BALANCER
2. COPY DNS NAME
3. PASTE IN BROWSER

[HTTP://3.82.48.11/](http://3.82.48.11/)



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